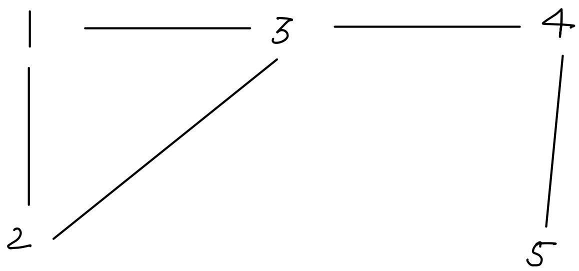
Curriculum

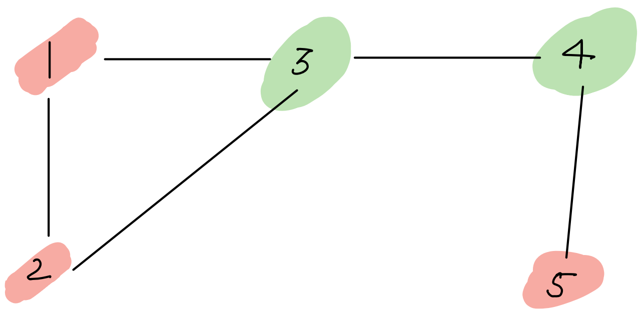
- Articulation pts & bridges
- AVL tree
- Binary lifting + LCA
- Combinatorics ★★
- DP
  - Digits
  - DP on trees
- Matrix exponentiation
- Nim Game.
- Segment trees [3]
- Tree flattening

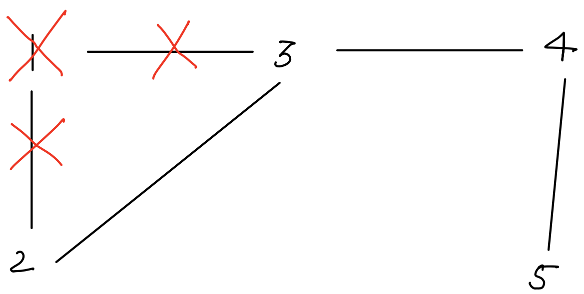
Articulation pts & bridges

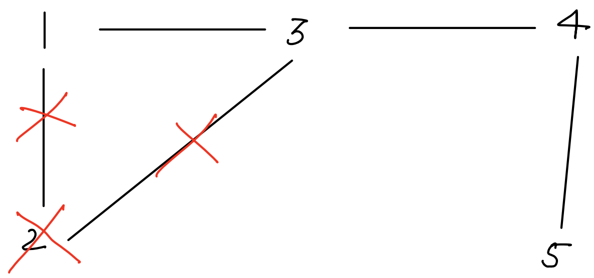


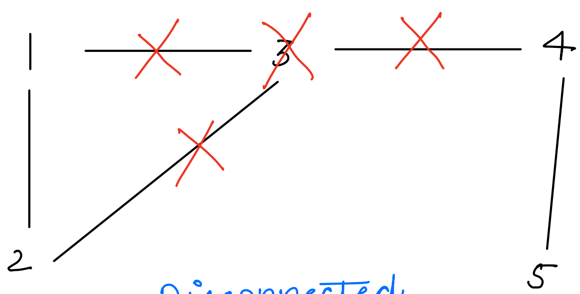
find articulation pt in this graph?

Nodes whose removal will make graph disconnected.









Disconnected

Brute force

~~bfs / dfs~~ for all nodes ——— $O(V)$

$O(1)$ — 1. Remove node from graph

$O(V+E)$ — 2. See graph is connected or not ?
$O(n)$

$O(1)$ 3. Add node.

TC: $O(\overset{n}{V} \cdot (\overbrace{V+E}^{n})) \simeq O(\overbrace{n^2}) =$
SC:

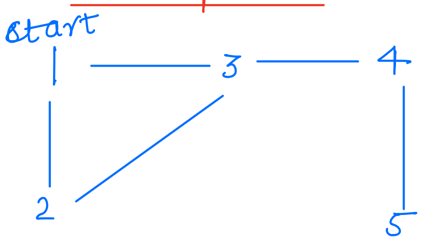
vertices

edges

$O(V \cdot (V+E))$

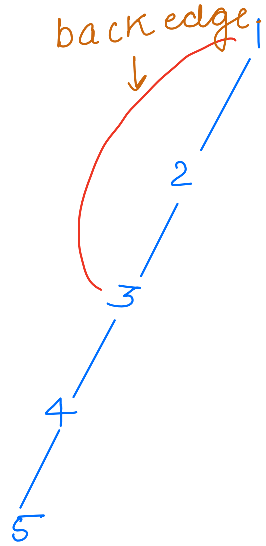
Back edges —— To find articulation pts

Graph

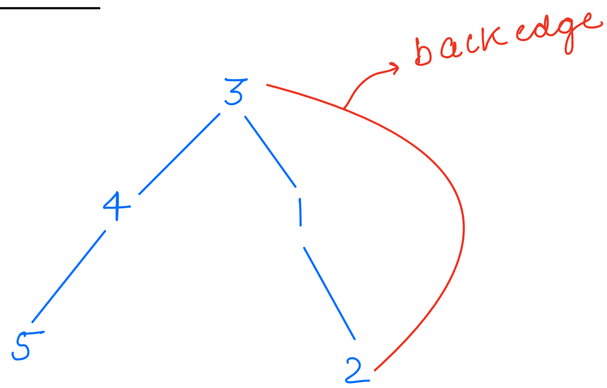


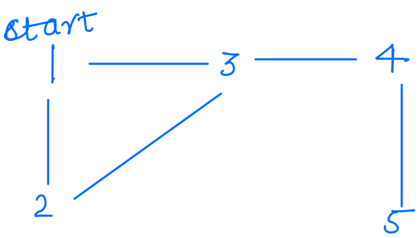
Visualation as tree

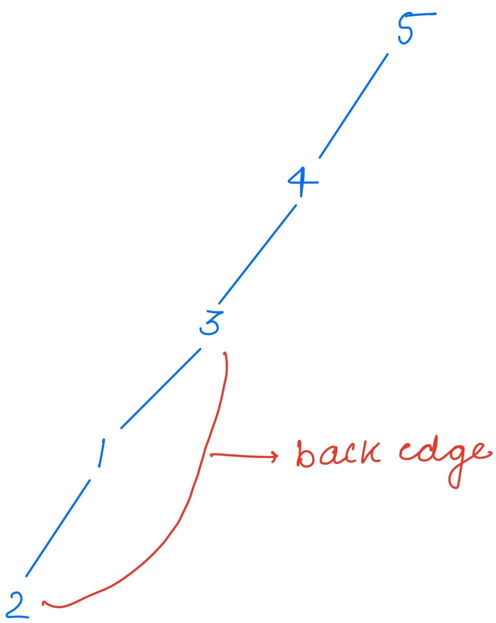
Dfs (1)



Dfs (3)

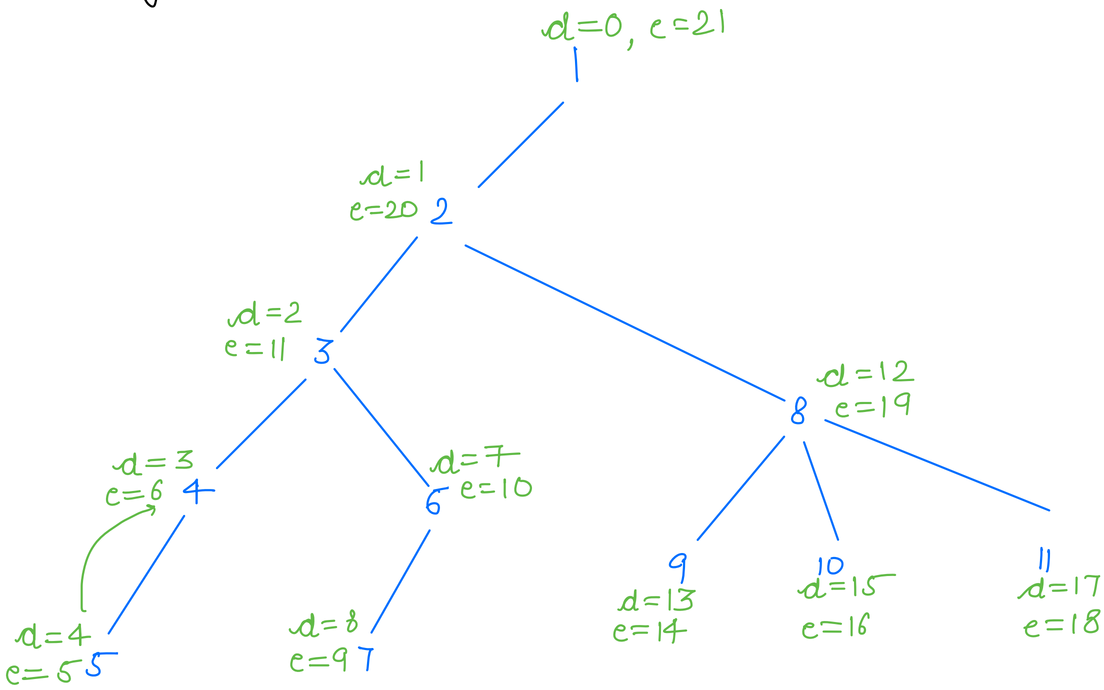






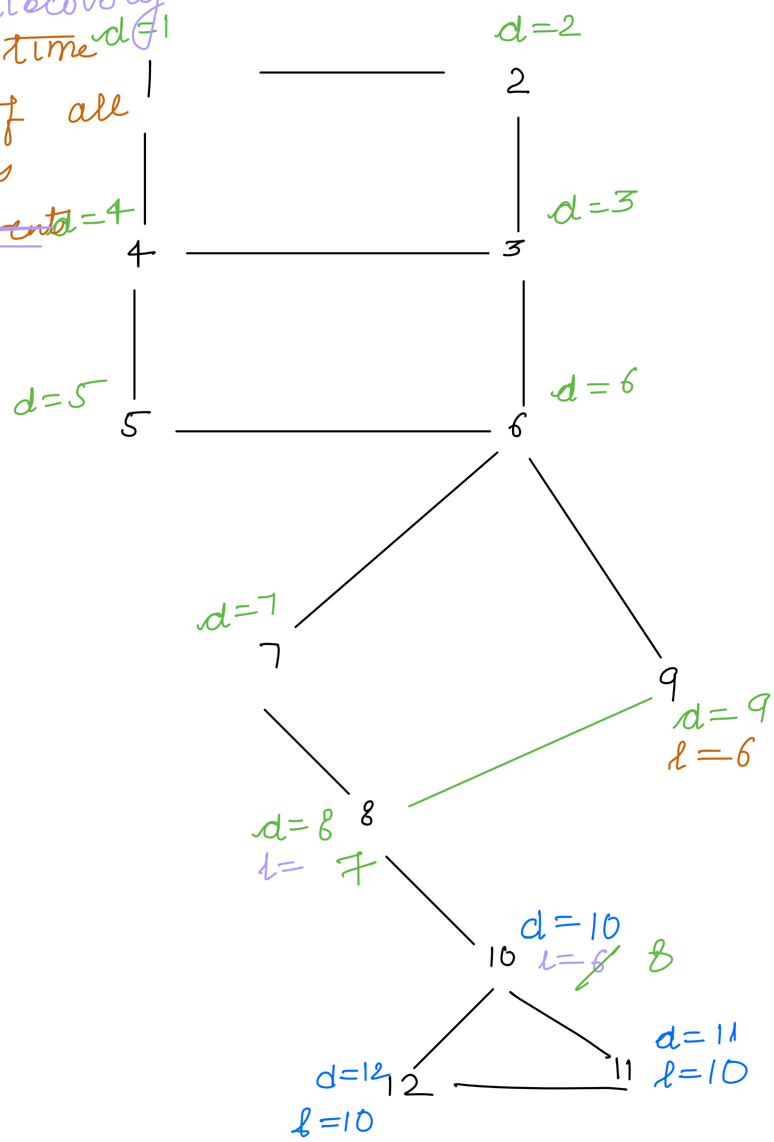
in/out time
discovery/end time

t=0



Lowest time

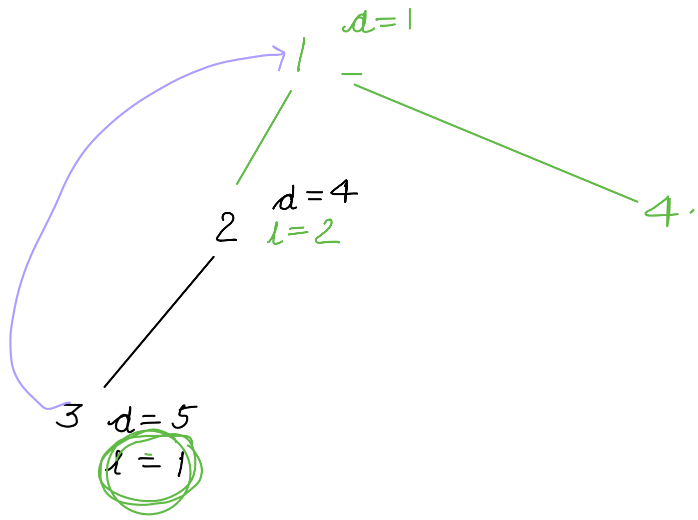
min lowest ~~discovery~~ time insertion of all adjacent node ~~apart from parent~~



if (lowest [child] < discovery[parent]) {

// child has a back-edge —— h/w

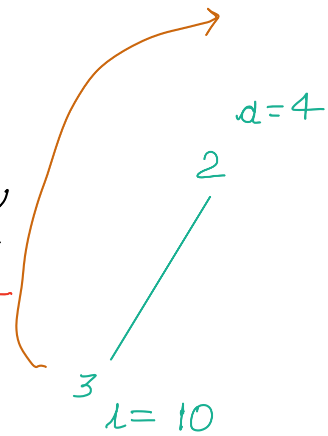
}

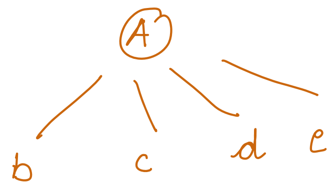


if (lowest [child] > discovery[parent]) {

// child does not has a back-edge

}

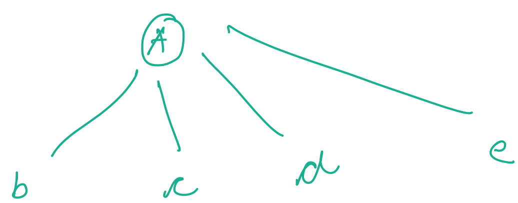




if any of child of A {b, c, d, e} does not have a backedge, I am AP

↑

A

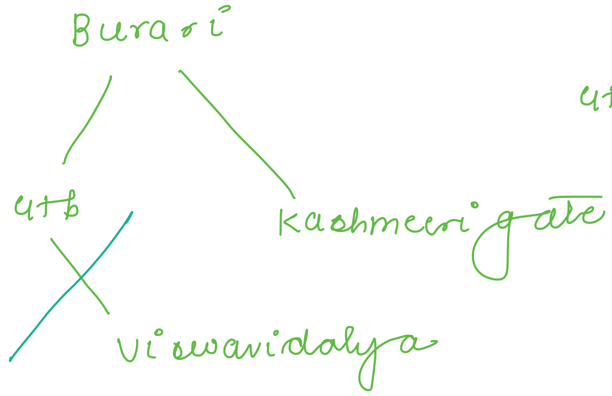


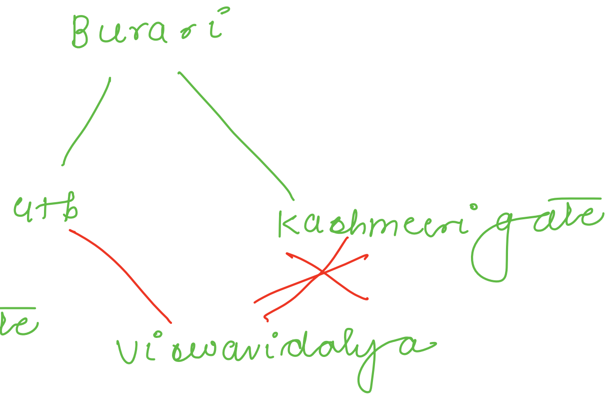
Break: 10:30 —— 10:40

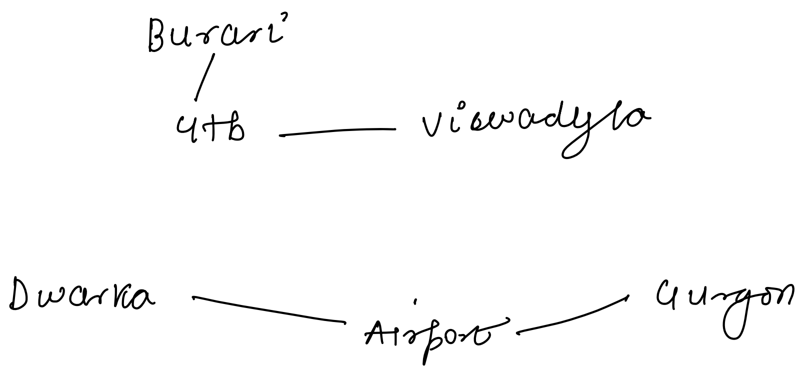
Destructive mind

Transport network
in form of graph

Q queries
each query $\longrightarrow$ node x.
whoever has a destructive mind
delete node x
we need to tell if city transport
is affected or not ?







if you have a disconnected graph, deleting a node/ edge, if it inc^r no of disconnected components, then only the city is affected.

Thankyou :)

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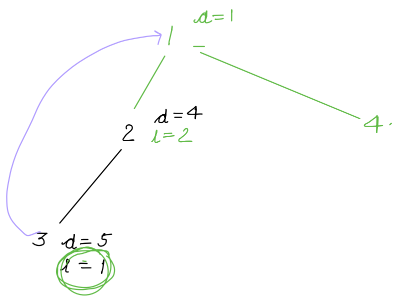
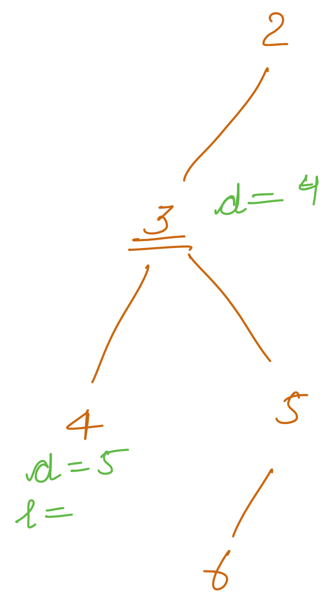
}





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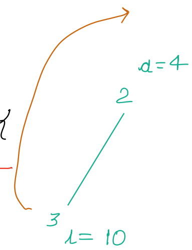

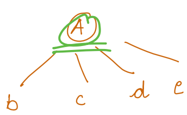

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A